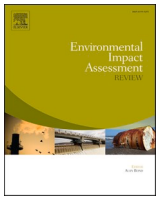
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Unveiling the anthropogenic dynamics of environmental change with the stochastic IRPAT model: A review of baselines and extensions☆

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ABSTRACT

While physical and natural scientists have long produced quantitative tools to understand biogeochemical processes, questions remain on how to identify, isolate and deliver unbiased measures which accurately shape the anthropogenic-global environmental change response function. This paper provides an in-depth review of IPAT, ImPACT, and STIRPAT baselines, combined with the most relevant theoretical extensions to those frameworks (e.g., ecological elasticities, *T*-term decomposition, and augmented Carbon Kuznets Curve functional form). Then, it offers a state-of-the-art survey of previous assessments combining STIRPAT settings with real time-series, stratifying by multi- and single-country approaches. We show how this literature failed to generate a strict consensus with respect to the magnitude of population and affluence elasticities, although some general principles seem to emerge. As part of an extensive methodological discussion, we next stress why those divergences can be linked to the variety of models' specifications and control variables' choices reported, but also the diverse manners through which past STIRPAT analyses attempted (or avoided) to address econometric challenges (e.g., unidentified simultaneity bias, use of multicollinear vectors and non-stationary data exhibiting a unit root; uncontrolled cross-sectional dependence and spatial correlation among the series; heterogeneous slope coefficients, non-linearities, asymmetries inferences; and lack of univariate and partial β-standardized parameters); and propose tools to empirically reconcile them. Finally, the (unlocked) policy potential of STIRPAT frameworks in unveiling competitive human-environment dynamics is stressed, whereas prospects for future research, with fruitful directions for both theory and empirics, are highlighted.

1. Introduction

While climate change has emerged as a major issue to be addressed, it is now admitted that anthropogenic activities modify the global environment. Over decades, the combustion of fossil fuels has dramatically modified the chemical equilibrium of the atmosphere by releasing Greenhouse Gases (GHG) and ozone particles (York et al., 2003). Thus, major biogeochemical cycles are being disrupted, whereas dramatic concerns are rising about the future of food security, biodiversity conservation and populations living in coastal zones. However, before implementing any comprehensive policy, unbiased measures of the marginal costs and benefits from environmental pollution are required (Deschenes and Meng, 2018). While physical and natural scientists created sophisticated tools modeling biogeochemical processes, questions long remained on how to determine, isolate and estimate the response of global environmental change to a set of different but identified anthropogenic factors. Over the past decades, researchers have therefore expressed a growing attention towards the establishment of a more reliable functional form modeling the relationship between anthropogenic drivers and environmental impacts; with *in fine*, the objective of delivering robust elasticities estimations for policy purpose.

A first stone was laid by Ehrlich and Holdren (1971) and Commoner et al. (1971) who provided a theoretical and algebraic foundation to the IPAT model, later enriched with additional policy leverages in Waggoner and Ausubel (2002), as part of the ImPACT formalization. While the former identity conventionally conceptualized total emissions (I) as the multiplicative product of population (P), per capita GDP (A) and CO_2 emissions per unit of GDP (T), the latter displays the advantageous policy feature of modeling CO_2 emissions (I) as the single outcome of population (P), per capita GDP (A), energy consumption per unit of GDP (C) and CO_2 emissions per energy consumption unit (T). Although both

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Nomenclature

2WFE	two-way Fixed Effects (cross-section and time)
AMG	Augmented Mean group estimator
ARDL	Autoregressive Distributed Lag Bound
BH-CC	Bayer-Hank Combined Cointegration
CCEMG	Common Correlated Effects Mean Group estimator
CIS	Commonwealth of Independent States countries
CKC	Carbon Kuznets Curve
DCCE	Dynamic Common Correlated Effects model
DiD	Difference-in-Differences
EE	Ecological Elasticity
EEO	Emission-Energy-Output
EKC	Environmental Kuznets Curve
FD-GMM	First-Differenced-Generalized Method of Moments
FDPR	First-Difference Panel Regression
FE	Fixed Effects
FE-DK	Fixed Effects with Driscoll-Kraay standard errors
FE-IV	Fixed Effects with Instrumental Variables
FGLS	Fully Generalized Least Squares
FMOLS	Fully Modified Ordinary Least Squares
GC-VECM	Granger Causality based on a Vector Error Correction Model
GMM	Generalized Method of Moments
IPAT	Impact by Population, Affluence and Technology
IVs	Instrumental Variables
KCT	Kao Cointegration Test
MG	Mean Group estimator
MM-QR	Method of Moments Quantile Regression
OLS	Ordinary Least Squares
PCSE	Panel-Corrected Standard Errors
PCT	Panel Cointegration test
PDCT	Pedroni Panel Cointegration test
PFE	Pooled Fixed Effects
PGC	Panel Granger Causality test
P-OLS	Pooled Ordinary Least Squares
PW-R	Prais-Winsten
QRM	Quantile Regression Model
RDD	Regression Discontinuity Design
RR	Ridge Regression model
SDM	System Dynamics Model
S-GMM	System Generalized Method of Moments
SO2	Sulfur Dioxide emissions
SP	Semi-Parametric Model
STIRPAT	Stochastic Impact by Regression on Population, Affluence and Technology
TRM	Threshold Regression Model
TSCS	Time-Series Cross-Section
VAR	Vector Auto-Regressive
VIF	Variance Inflation Factor
WEHS	World Economy Historical Statistics

IPAT and Impact present a competitive grounding in ecological principles captured by a simple but pretty clear equation specification, they shortly exhibited theoretical and empirical limitations. Upon them, they failed to allow hypothesis testing because the accounting structure of those equations takes the form of a product of several factors. Hence, the identified values of some variables influence the unknown content of the missing term (York et al., 2003). Also, homogeneous unity coefficients weighted the relative importance of each factor within the specification, which conflicts with the socio-ecological theory in search of non-proportional dynamics principles characterizing the human-environment nexus. Finally, they did not capture the non-linear and non-monotonic shape of the anthropogenic activity environment response function, despite its substantial place given by the Environmental Kuznets Curve (EKC) theory (Grossman and Krueger, 1991; Panayotou, 1997; Stern, 2004). Instead, the scope of those models was constrained because of the underlying assumption that a doubling in the magnitude of a given variable should induce a doubling of impacts, holding other drivers constant (York et al., 2003).

Facing those caveats, Dietz and Rosa (1994) offered a stochastic extension of this framework which allows for more flexible and rigorous assumption testing. Being no longer an accounting identity, the Stochastic Impact by Regression on Population, Affluence and Technology (STIRPAT) model revolutionized the empirical viewpoint to environmental impact assessments because it encompasses a constant scaling the overall model, individual exponents for each variable, and an error term all-together. Combined with a novel non-linear structure, it enabled researchers to engage with sociological theory and assess the relevance of their hypotheses (i.e., EKC, CKC) with real time series data (McGee et al., 2015). While estimated parameters remain multiplicative functions of P, T and A, they can be individually estimated and decomposed into sub-components. In doing so, new influencing factors could be considered following each case study's specificity, thus offering a high degree of extension potential to the benchmark framework (Wu et al., 2021). From the concept of Ecological Elasticities proposed in York et al. (2003), to the decomposition of T (Elliott and Clement, 2014), *antilog* of the residual conventionally included within the error term when not estimated separately and which models technology as the net out of all other factors determining air, water, soil and biodiversity quality changes; seminal studies enriched the STIRPAT modeling literature; which later reconciled Carbon Kuznets Curve (CKC) and Emission-Energy-Output (EEO) analyses as part of a unique framework (Liddle, 2013a).

But interestingly, the range of multi- and single-country empirical assessments which followed the development of those conceptual models failed to generate a consensus with respect to the nature and shape of the human-environment response function. While case studies, data periods, and methodological approaches differ, the sign and magnitude of population ($\widehat{\sigma}_{IP}$) and affluence ($\widehat{\sigma}_{IA}$) elasticities seem to have diverged rather than converged, thus making policy implications less reliable. Naturally, the multi-faced nature of T encompassing any element influencing per unit production reduces the potential for estimating technology. Instead, it opens the door to a wide range of different combinations of control variables and lowers the generalization potential of empirical estimates. But another reason may lay in the multitude of econometric challenges arising from previous empirical STIRPAT analyses, which they sometimes attempted to partially address. Whether it concerns identification strategies which exclude multicollinear vectors, reverse causality and simultaneity bias, regression estimations performed on transformed non-stationary data exhibiting unit root processes, uncontrolled cross-sectional dependence among the series and spatial-correlation inferences, partially accounting for heterogeneous slopes within the sample by estimate a unique vector of parameters for each country, failing to allow for non-linearities and asymmetries in the estimated responses, or omitting to open a potential for univariate and partial β-standardizations of coefficients with direct comparability benefits across studies; those deserve an explicit review, investigation, and discussion.

For the aforementioned reasons, there is a point in providing an exhaustive analysis of past and present theoretical STIRPAT models set-up, combined with accurate methodological and empirical takeaways, as well as directions left to the domain for future research. Accordingly, this study seeks to extend the current literature in four distinct manners:

- *First*, this paper is the first to provide an in-depth review of IPAT, ImPACT, and STIRPAT baselines, combined with the most relevant theoretical extensions to those frameworks (e.g., ecological elasticities, T -term decomposition, and augmented Carbon Kuznets Curve functional form) (*i*).
- *Second*, this study contrasts to previous ones as it offers a state-of-the-art survey of previous assessments combining STIRPAT settings with real time-series, stratifying by multi- and single-country approaches, along with a critical note about studies' characteristics, conflicting conclusions, and the emergence of some general principles (*ii*).
- *Third*, this research displays a competitive edge as it supplies an extensive methodological discussion stressing how those divergences can be linked to the variety of models' specifications and control variables' choices reported, but also the diverse manners through which past STIRPAT analyses attempted (or avoided) to address a set of econometrics challenges, with tools to attempt a reconciliation (*iii*).
- *Finally*, we stress the unlocked policy potential of STIRPAT frameworks in unveiling the anthropogenic dynamics of environmental change, and thus highlight prospects for future research, with fruitful directions for both theory and empirics. Filling this gap is where our study finds its fourth and last contribution (*iv*).

Besides the first Section, this paper is organized as follows. Section 2 introduces the foundations of the IPAT and Impact models. In Section 3, the model set-up of the STIRPAT model is presented, whereas Section 4 presents its recent extensions. In Section 5, a state-of-the-art review of empirical STIRPAT assessments is displayed whereas in Section 6, an extensive econometric discussions is offered. Section 7 offers prospects for future research while concluding remarks are presented in Section 8.

2. Background: IPAT and ImPACT models

The origin of the IPAT (Impact by Population, Affluence and Technology) model can be traced back to Ehrlich and Holdren (1971). They pointed out that Coale (1970) failed to estimate the effective contribution of the population multiplier and thus had underestimated how vary the population-driven resource demand and associated environmental impacts. Hence, they developed the $I = P \times A \times T$ functional form, and which led to its current definition. Followingly, Commoner et al. (1971) provided a theoretical and algebraic foundation to this concept, as well as the empirical examination using data series. However, a debate arose as this latter stressed that the leading channel through which anthropogenic activities modify the environment is technological change instead of population dynamics. The main question laying under the intense Ehrlich–Holdren/Commoner debate became whether and how increasing population improve technologies; or reciprocally, does increased technology trigger carrying capacities (McGee et al., 2015). IPAT has since been employed to model environmental change as the multiplicative outcome of three factors: population (i.e., demographic force), affluence (i.e., per capita production or consumption of goods and services) and technology (i.e., impact per unit of consumption or production). Therefore, one should interpret $I = I \times P \times A \times T$ as a mathematical identity representing an accounting function, in which identified values of *I*, *P*, and *A* are utilized to solve for *T* using a standard equation-solving approach (Raskin, 1995; York et al., 2003). Since affluence is typically used as a proxy for per capita Gross Domestic Product (GDP), *T* conventionally captures the environmental impact weighted per unit of production, such that:

$$P \times A = P \times (GDP/P) \Leftrightarrow T = I/(PA) \Leftrightarrow T = I/GDP \quad (1)$$

The IPAT model presents several advantageous features. First, it identifies and estimates the environmental elasticity to changes in a set of distinct driving factors. Second, the functional form allows researchers to assume that all variables (*P*, *A* and *T*) are unlikely to generate changes independently from one another. In other words, because factors interact through a product multiplying them all together, this prevents one particular factor for being held as singularly responsible for a given environmental change. Instead, one should interpret those dynamics as containing a scale effect induced by the inter-relationships among variables. For instance, York et al. (2002) show that if for a given country *i*, *P* and *T* do not rise or decline over a given time period whereas *A* records an increase, this becomes inconsistent to conclude that *A* is the sole driver of environmental impacts, because *P* and *T*, although they stay constant, influence the scale of the impact induced by rise in affluence. Hence, discerning the relative importance of those individual factors cannot go without examining the slope and set of potential variation in each of them. For this reason, York et al. (2002) proposed and developed the pioneer concept of plasticity which measures the potential variation ranges in population, affluence, and technology-driven environmental responses. In doing so, these authors modeled how factors generate impacts differing in their scales and rates of occurrence by distinguishing between the potential range and variability of each factor (i.e.,*"the potential breadth of change"*) (*i*); and the rate at which they can change (i.e.,*"the potential speed of change"*) (*ii*) (York et al., 2003). One novelty brought by this concept is the substitution potential of those factors, although questions remained on which factor, when regulated, embeds the highest policy impact and efficiency.

To fill this gap, Waggoner and Ausubel (2002) extended the baseline of the IPAT model with additional policy GDP unit starting from the benchmark setting, they divided *T* into consumption per unit of GDP (*C*) and impact per unit consumed (*T*), leading to the following revised form $I = P \times A \times C \times T$, and named ImPACT. For instance, if one models carbon dioxide (CO_2) emissions as predicted by a set of anthropogenic drivers, the conventional IPAT identify would conceptualize emissions (*I*) as the inter-connected product of population (*P*), per capita GDP (*A*), and CO_2 emissions per unit of GDP (*T*). Conversely, the ImPACT framework presents the advantageous feature of modeling total CO_2 emissions (*I*) as product of population (*P*), per capita GDP (*A*), energy consumption per unit of GDP (*C*) and CO_2 emissions per unit of energy consumed (*T*). Hence, the ImPACT model displayed an incipient competitive edge for policy purpose as it helped identifying the factors that can be mitigated to reduce global environmental impacts (York et al., 2003).

While attempting to identify additional factors affecting the environment, Diesendorf (2002) introduced the $I = P \times B \times A \times T$ model, where *B* accounts for behavior. However, Diesendorf (2002) provided three reasons against multiplying the right side of the IPAT equation. First, behavioral dynamics were implicitly but already present within the benchmark model because variations in total population, population-weighted consumption and technological development are partly determined by decision-making processes relevant to behaviours. Thus, including *B* is likely to generate bias because it double-counts the weight taken by this force. Second, Schulze (2002) did not precisely define the multi-faced features characterizing an arbitrary behaviours variable, and which empirical metrics and data series could capture those elements in a single manner. Third, the accounting identity $I = P \times A \times T$ results from a non-arbitrary selection of factors since the disaggregation of *I* into *PAT* is rationally and mathematically meaningful, unlike *PBAT*. Assuming the formula $I = I$; multiplying its right-side by *CP/CP*; with *C* refers to absolute consumption or GDP, gives the logical:

$$I = (I/C) \times (C/P) \times P \Leftrightarrow I = P \times A \times T \quad (2)$$

where the second-side of this equation can be obtained when assuming $T = I/C$ and $A = C/P$, and inverting the order of the three factors. However, Diesendorf (2002) showed that modifying Eq. (2) to an equality that can be resolved for an undetermined impact *T* requires to redefine this identity, independently from *I*. Hence, *A* would have to be re-framed as well, which is inconvenient because it embeds the ecological impact of GDP; and reduces the policy implications derived

from the interpretation of a GDP-free specification; as shown in Diesendorf (2002):

$$I_{ghg} = \left(I_{ghg}/E\right) \times (E/P) \times P \Leftrightarrow I_{ghg} = P \times A_{ghg} \times T_{ghg} \quad (3)$$

where I_{ghg} captures the environmental impacts from GHGs emissions, sourced by energy consumption E. The left-hand side of the above functional form is obtained by multiplying I_{ghg} by EP/EP; and setting $A_{ghg} = E/P$ as a energy-based measure of affluence (i.e., energy consumed per unit of population; and $T_{ghg} = I_{ghg}/E$ as a energy-based measure of technology (i.e., GHGs emissions per unit of energy consumed). For an more exhaustive discussion on this topic, see Roca (2002).

Overall, it is widely admitted that IPAT and ImPACT display a competitive grounding in ecological principles, relevantly captured by a simple but clear equation specification. They were among the first models to explicitly illustrate how changes in anthropogenic factors can modify environmental impacts. Nonetheless, as pointed out in York et al. (2003), they present the disadvantageous features of removing the possibility of any hypothesis testing. The reason is that those accounting equations specify a product of several factors. Thus, established values of some variables influence the unknown value of the missing element. Additionally, homogeneous unity coefficients weight the relative importance of each factor within the specification, which is a critical limitation as assuming proportionality in the respective impact of each driver may conflict with the socio-ecological theory. Hence, IPAT and ImPACT fail to capture the non-linear and non-monotonic shape of the anthropogenic activity-environment response function, nonetheless frequently highlighted in the Environmental Kuznets Curve (EKC) literature (Grossman and Krueger, 1991; Panayotou, 1997; Stern, 2004). Besides, affluence (A) is likely to exhibit a non-proportional effect on environmental quality indicators, thus giving the final and well-established inverted U-shaped polynomial form of the curve. Instead, the scope of those models is constrained because they assume that a doubling in the magnitude of a given variable should induce a doubling of impacts, ceteris paribus. Widely recognized as the founding framework of human-environment nexus assessments, IPAT and ImPACT identities failed to account for non-monotonic and non-proportional dynamics in the above relationship. This is where the stochastic extension finds its contribution.

3. The STIRPAT framework

Dietz and Rosa (1994) reformulated the IPAT model by introducing a stochastic feature allowing for more flexible and rigorous assumption testing and empirical estimation. It differs with previous frameworks as it does not take the form of an accounting equation. Instead, the general functional-form of the Stochastic Impact by Regression on Population, Affluence and Technology (STIRPAT) model includes individual exponents for each variable as well as an error term, such that:

$$I_{i,t} = c_t \times P_{i,t}^{\alpha} \times A_{i,t}^{\beta} \times T_{i,t}^{\gamma} \times \varepsilon_{i,t} \quad (4)$$

where the constant c intends to scale the model according to the different dimensions and metrics of variables, whereas α, β, and γ are the exponents of P, A, and T, respectively, and refer to the elements which are to be estimated. Finally, ε refers to the error term. Finally, subscripts i and t indicate that I, P, A, T, and e quantities change across units (countries, provinces or regions) and time periods. Notice that IPAT's proportionallity hypothesis assumes $1 = a = b = c = d = e$ (Dietz and Rosa, 1994). Taking the logarithmic transformation allows much easier panel estimation procedures and hypothesis testing assessments. Estimated paramaters can be interpreted as marginal ecological elasticities in which impacts are multiplicative products of population, technology, and affluence (McGee et al., 2015). Notice that in the benchmark STIRPAT formulation, T is comprised within the error term, and not regressed separately because in the IPAT identity, T is solved for I, P and A. A log-transformation functional form can be expressed as:

$$log\left[I_{i,t}\right] = ln[c_t] + \alpha\left(log\left[P_{i,t}\right]\right) + \beta\left(log\left[A_{i,t}\right]\right) + \gamma\left(log\left[T_{i,t}\right]\right) + \varepsilon \quad (5)$$

where α, β, and γ capture individual elasticity from a given environmental metric to population, affluence and technology, respectively. One sees that Eq. (5) can no longer be considered as an accounting identity with right and left side dimensions which must balance, but becomes a more flexible tool for hypotheses testing with data series (Liddle, 2015). Often, empirical studies pointed out that panel data turn integrated of order 1 ($I(1)$), indicating the presence of a unit root in level-series, but rejection of the null hypothesis of non-stationarity when first-differenced-transformed (Martínez-Zarzoso and Maruotti, 2011; Sadorsky, 2014; Usman and Hammar, 2021). Therefore, most STIRPAT frameworks have been adapted to incorporate the first-difference structure of the series. Those present the advantage of assuming constant unobserved unit-effects and accounts for cross-sectional bias (Wooldridge, 2015). The first-difference-log-transformed STIRPAT specification becomes:

$$\Delta log\left[I_{i,t}\right] = \Delta log[c_t] + \alpha\Delta log\left[P_{i,t}\right] + \beta\Delta log\left[A_{i,t}\right] + \gamma\Delta log\left[T_{i,t}\right] + \varepsilon \quad (6)$$

One appealing feature of the Stochastic computation of the IPAT model is its ability to better identify the dynamics coupling human systems and environmental ecosystems. By allowing for non-monotonic response functions between drivers and impacts (i.e., EKC), it allows researchers to engage with sociological theory and assess the relevance of their hypotheses with real time series data (McGee et al., 2015). It does not only allow each coefficient to be estimated as individual parameters, but also enables a decomposition analysis of each factor. Hence, the standard STIRPAT framework can be extended with new influencing factors based on case studies' characteristics (Wu et al., 2021). Also, Jorgenson and Clark (2010) showed that this tractable regression equation dampens the skewed distribution of the variables. In doing so, it corroborates Schulze (2002)'s intuition stressing that some factors different than P, A, and T may explain environmental change, and should thus be incorporated. However, the standard specification as represented in Eq. (4) assumes homogeneous elasticities for all vectors of observational units i, which avoids the possibility of structural differences across countries and regions. Later, Pesaran (2006) filled this gap by offering the architecture of the Common Correlated Effects Mean Group (CCEMG) estimator which accounts for heterogeneous slope coefficients and cross-sectional dependence among units.

All in all, *IPAT/STIRPAT* identities became the most classical theoretical frameworks to examine the drivers of environmental pollution, and especially atmospheric indicators. They also planted the seeds of an entire empirical segment of the literature whose methodological features are presented in Section 5.

4. Theoretical extensions of IPAT and STIRPAT models

This Section aims at presenting the most relevant theoretical extensions of IPAT and STIRPAT identities, which also refer to the seminal and pioneer novelties offered in Liddle (2015); McGee et al. (2015); York et al. (2003).

4.1. Introducing the concept of ecological elasticities

York et al. (2003) refined the STIRPAT model to become an analytical tool and extended its potential for empirical applications. To do so, they developed the concept Ecological Elasticity (EE), in line with their introductory concept on *plasticity* discussed previously. While the latter corresponds to the possibility that P, A, and T could endogenously vary, the former connects with the sensitivity/response of the environment to those variations in the driving forces. In economics, the general concept of elasticity is defined as a measure of the responsiveness of the

dependent variable to individual changes in another explanatory one, with other factors held constant (i.e., ceteris paribus) (Samuelson and Nordhaus, 1992). This presents the advantageous feature of offering a percentage-based (%) response to a unity percentage change (1%) in a given variable, which gives a broader picture on how sensitive is the environment to anthropogenic dynamics. York et al. (2003) started from the conventional price (*P*) elasticity specification, as a function of the quantity supplied (*Q*):

$$log[Q] = log[A] + b \times log[P] \quad (7)$$

where b captures the price elasticity of supply whereas $log[A]$ is a constant scaling the overall model. When a residual ε term is added at the end of the equation, Eq. (7) ends up becoming a stochastic identity enabling statistical estimations of the $\widehat{b}$ coefficient with consecutive observations over time, or cross-sectional data (York et al., 2003). Therefore, one sees that the stochastic extension of Eq. (7) is functionally equivalent to the STIRPAT framework specified in Eq. (5). Having stated this information, York et al. (2003) defined EE as capturing the proportional response of environmental changes due to a marginal variation in any anthropogenic variable, or alternatively, EE refers to individual component elasticities associated with each driver identified in the STIRPAT equation. Accordingly:

- EE_{IP} refers to the elasticity of impact to population, which is the environmental change sensitivity to a marginal variation in the total size of the population (i.e., expressed as a continuous nation- or region-wide variable), *holding all other variables constant*.
- EE_{IA} refers to the elasticity of impact to affluence, which is the environmental change sensitivity to a marginal variation in the economic metric capturing affluence (i.e., per capita GDP expressed as a continuous nation- or region-wide variable), *holding all other variables constant*.
- Estimated parameters displaying positive magnitudes $m > 1.0$, $0 < m < 1.0$, $m = 1.0$ indicate a more than proportional (growth rate of environmental change > growth rate of the driving force), inelastic (growth rate of environmental change < growth rate of the driving force) and a unit proportional relationship (growth rate of environmental change = growth rate of the driving force), respectively.
- Conversely, $m = -1.0$ corresponds to a negative unity elasticity (environmental change decreases proportionately to an increase in the anthropogenic force); whereas $m < -1.0$ shows that environmental impacts decline more than proportionately than a unitary increase in the driving force. Overall, $-1.0 < m < 0$ reveals that the rate of environmental impacts decline turns smaller than the growth rate of the driving force.

Naturally, York et al. (2003) did not develop the elasticity of environmental change to technology because, as discussed in the next Sub-Section, *T* is a multi-faced aggregated variable encompassing several factors (any element influencing per unit productivity). That is, while it does not prevent us from estimating its associated elasticity, interpreting the policy meaning of this statistical measure without disaggregation turns out to be much more challenging. In general, combining the concept of Ecological Elasticity (EE) with a STIRPAT framework allows the inclusion of additional factors within a multivariate setting, and thus control for possible endogeneity bias induced by omitted variable correlated with the residual term ε. Nonetheless, the multiplicative structure of the STIRPAT model should not be omitted. Instead, considering new potential covariates can be conceptualized as adding sub-components of the aggregated variable *T*, as long as they are thought to possibly confound the population- or affluence-environmental change nexuses (York et al., 2003).

4.2. Proposing a disaggregated version of T

In the standard STIRPAT approach, *T* is incorporated within the residual term ε, and not estimated separately as this latter becomes resolved for known *I*, *A*, and *P* variables. Thus, when technology is not proxied by a mixture of variables (i.e., energy intensity or R&D expenditure), it is often considered as the *antilog* of the residual, which models technology as the net out of all other factors driving environmental change (i.e., encompassing any element influencing per unit production), aside from population (*P*) and affluence (*A*) (Dietz and Rosa, 1997; Dietz et al., 2012; York et al., 2003). Given the multi-faced nature of the technology (*T*) parameter, analyses progressively attempted to disaggregate *T* into several sub-factors in the original IPAT identity (which is equivalent to disaggregating *CT* in the ImPACT identity), thus rendering technology more operationally flexible in its empirical uses and purposes.

While such decomposition concept does naturally apply to *P* and *A*,[1] a long line of socio-ecologists suggest that anthropogenic drivers different from population and affluence influence global environmental change. For instance, the ImPACT identity exhibits consumption as an additional factor. Hence, as long as those variables are deemed appropriate to capture an omitted variable correlated with the error term ε, and thus partially determine environmental change, they can no longer be neglected. Nonetheless, considering political variables (e.g., quality of political regime, index of cultural integration and preferences, but also environmental treaty ratification) turns limited since those inferences are typically captured by non-continuous string variables with multiple categories (i.e., nominal or ordinal measurements) which do not play out in a multiplicative framework (Frank et al., 2000). Instead, dummy coded (0–1) variables may be used, although the binary nature of these data prevents from considering three categories and more. Hence, generating bins corresponding to identified quantiles could partially fill this gap, as proposed by York et al. (2003). Those authors worth noticed that dummy coded covariates prevent from elasticity coefficient interpretations. Instead, the antilog (exp_x) of the parameter for any dummy element should be seen as the multiplier linking the unity coded value of the dummy with respect to the null (i.e., unity of event against the null of no event). All in all, the STIRPAT functional form presents the advantageous feature of enabling researchers to estimate sub-components elasticities of *P*, *A* and *T*, or their initial fully-aggregated structures. Among the most relevant empirical decomposition of *T*, Shi (2003) relied proxied technology influence with industrial value-added as a percentage of GDP and services value-added as a percentage of GDP; whereas Wang et al. (2013) employed energy intensity per unit of total output. Finally, Roberts (2011) considered urbanization (proxied as the share of urban population and the proportion of constructions in a metropolitan area) while McGee et al. (2015) extended this literature in connecting this framework to impervious surface areas. By operationalizing technology as terrestrial development, they provided a reliable setting in which it can be framed independently from *P* and *A*. In doing so, they reconciled STIRPAT and urban sociology litteratures over the anthropogenic construction of the built environment (Elliott and Clement, 2014; Logan and Molotch, 2007).

For IPAT and ImPACT models, examining carbon intensity (but it also applied to efficiency) measure is allowed by $T = I/(P \times A)$; where *I* indicates CO_2 emissions. However, it fails to be an independent intensity measure of *T*; and turns instead to take the form of a value imbalancing the IPAT identity.[2] Thus, York et al. (2003) argued that the estimated

[1] Cramer (1998) was the first to distinguish between the number of households and average household size when estimating air quality sensitivity in California; whereas York et al. (2003) used the percentage of the population in economically productive age categories.

[2] York et al. (2003) suggested that only a ratio measure of *T* derived from the other terms may re-balance the identity

measure of T might neither capture a consistent cross-unit variation in intensity, nor the true intensity reduction potential. A set of potential independent measures of technologies end up excluded, whereas IPAT coefficients α and β (i.e., equivalent to $log[I] = c + \alpha \times log[P] + \beta \times log[A] + \varepsilon$) derived from a non-linear specification may capture a combination of linear-, quadratic-, cubic- or/and higher polynomial structures of degree n, where the intensity formula takes a more flexible and accurate form: $T = I/[c \times P^{\alpha} \times A^{\beta}]$. Thus, the only way to account for an individual change P or A that generates a neutral effect on I (and control for the multiplicative structure of this accounting equation) is to postulate that a variation in T with equivalent magnitude but opposite sign. While this is another novelty of the accounting nature of the equation, it turns out to become a particular case of the STIRPAT identity (i.e., when $EE_{IA} = 1.0$). In other words, this indicates that the effect per unit of production (T) does not vary as affluence increases; whereas $EE_{IA} > 1.0$ shows that environmental impacts grow faster than affluence. However, capturing this dynamic with a IPAT functional form, one must rely on the constraining assumption that the per unit of production effect (T) increases at the speed of affluence rise ($EE_{IA} < 0.0$), enough to ensure an absolute decline in environmental impacts net of positive inferences. Thus, when EE_{IA} appears not unit elastic, relying on T to re-balance the IPAT equation for variations in A is the only way to estimate the individual response of a variable that is free of multiplicative effects with others. Conversely, EE_{IA} measures offered by STIRPAT provides more reliable information about T, because the value exhibited by any variable in the identity can freely change without spurious co-determination by the values of the other terms. Hence, York et al. (2003) stated that it does not only enable the application of relevant association testing framework with more robust identification strategies, but also allows a better hypothesis testing with fewer falsification bias, and alternative functional specifications comprising polynomial forms, to infer causality, and *in fine*, permits projections of future impacts based on individual estimated coefficients.

4.3. Extending STIRPAT frameworks to EKC/CKC analyses

Another strand of the literature attempted to bridge Environmental (*EKC*)/Carbon Kuznets Curve (*CKC*) models with the STIRPAT framework. While it argues that carbon dioxide emissions initially increase with economic expansion, reducing afterward the turning point, whereas further economic activity leads to a decline in carbon emissions levels. Thus, this postulates that economic activity may exhibit an inverted U shaped relationship with carbon emissions. But beyond that, this de-linking dynamic underlines the encouraging plausibility that economic and environmental targets may be simultaneously achievable[3] Outlined by Lieb (2003), several economic mechanisms may theoretically explain the emergence of a threshold level in the income-pollution response function. This comprises a gradual demand for environmental goods and quality in general, internalizations of externalities within market operations, substitution effects between pollutants, low-carbon technology deployment, increasing returns to scale in pollution abatements, structural changes in the carbon intensity of aggregate income, migration/outsourcing of the most energy and resource intensive industries with carbon leakages with more lenient host countries. Referring to the STIRPAT framework as a landmark, *EKC/CKC* functional form is expressed as:

$$ln(I/P)_{it} = \alpha_i + \gamma_t + \beta_1 ln(A)_{it} + \beta_2 ln((A))^2_{it} + \beta_3 ln(Z)_{it} + \varepsilon_{it} \quad (8)$$

where α and γ refer to two-way (cross-sectional and time) fixed effects; whereas Z can be interpreted as a vector of other tested drivers (similar to T in Eqs. (4), (5) and (6); which is sometimes but not systematically considered). The *EKC* hypothesis is said to be empirically validated if and only if $\widehat{\beta}_1$ and $\widehat{\beta}_2$ are positive and negative statistically significant parameters, respectively. The *turning point* (i.e., the optimal level of output at which the income-atmospheric emissions nexus shifts from a positive to a negative pattern) can be determined by differentiating the estimated function with respect to the economic variable (*i*), equalizing the equation to zero (*ii*), and resolving for the unknown term (*iii*). Although bringing a polynomial structure to the STIRPAT functional form finds a theoretical support, York et al. (2003) argued that it may complicate the interpretation of Ecological Elasticity (EE) coefficients. On the one hand, the straightforward measure of elasticity/inelasticity over the linear term does no longer apply. Instead, elasticities derived from both quadratic and non-quadratic terms should be taken into account. However, the instantaneous elasticity coefficient can be calculated using the first partial derivative of the equation specification with reference to the logarithmic transformation (i.e., for percentage-elasticity interpretation purpose) of a given driving force. Looking at the quadratic model presented in Eq. (8), the affluence ecological elasticity of environmental impacts (EE_{IA}), for any value taken by $log[A]$ and holding $log[P]$ unchanged equalized $EE_{IA} = c + 2d \times log[A]$. On the other hand, expanding a STIRPAT with interaction term may artificially increase the variance of the model due to overidentification bias, rather than capturing unaddressed non-linearities.[4] The reason is that interactions are by definition multiplicative effects, which are indeed already captured by the structural form of the STIRPAT model (see Sections 3 and 4) (Dietz and Rosa, 1997).

While EKC and CKC literatures are rich and extensive, identification strategies in econometrics differ, and results often conflict.[5] Besides, Richmond and Kaufmann (2006) showed that if the estimated *threshold level* is found to locate out of the sample range, the final response function is more likely to take a semi-log or log–log form, instead of an inverted U shaped one. But because many EKC analyses failed to report implied turning points, the lack of consensus on this topic seems far from being over. One of the main explanations for this non-monotonic behavior the elasticity of environmental quality demand to endogenous changes in output must exceed unity (Dinda, 2004). Therefore, Liddle (2013a) pointed out an inverted U-shaped curve tends to more frequently emerge for pollutants whose adverse environmental effects are explicitly identified, adequately interpreted, with timely observed effects (smaller than a decade- or life-span), so that they can be controlled locally and monitored. As a matter of fact, if people affected by the public good are unable to assess, locate and implement direct policy responses to lower emissions, a turning point of GDP is unlikely to occur. This reasoning applies relevantly to environmental damages covering a large and global scale, with time lags, uncertainties and difficulties to be directly identified and witnessed by human societies (i.e., warming temperatures, ocean acidification, ice thawed). Over 2010's, *CKC* and energy-GDP nexus analyses attempted to be unified in a single testing framework taking some (but not all) features from the STIRPAT setting. Studies incorporated energy use as an additional independant contributing factor to the general *CKC* specification, thus creating the Emission-Energy-Output (*EEO*) model. (Apergis and Payne, 2009, 2010a; Lean and Smyth, 2010). The panel level-specification takes the following form:

$$C_{it} = a + bE_{it} + cY_{it} + d(Y)^2_{it} + u_{it} \quad (9)$$

[3] The basic philosophy of the EKC-theory is summarized in Beckerman (1992)'s viewpoint: *"although economic growth usually leads to environmental deterioration in the early stages of the process, in the end, the best and probably the only-way to attain a decent environment in most countries is to become rich" (p.482)*.

[4] See (Dietz and Rosa, 1997) proposed an exhaustive (and still up-to-date) discussion on the potential of interaction terms combined with a discussion on statistical limits.

[5] See Ben Jebli et al. (2022) for an exhaustive review of the EKC literature.

where C refers to carbon dioxide emissions (CO_2) per capita; E indicates population-weighted energy consumption; and Y indicates real GDP per capita; with u the disturbance term. *CKC* and STIRPAT models differ in the sense that the former one assumes unity for population elasticity whereas the latter does not. Hence, the *CKC* model transforms the dependent variable into a per capita formulation for each country i and period t (i.e., $ln(I/p)_{it}$) (Liddle, 2015). On the one hand, if the elasticity of population differs from one, then the standard representation of the *CKC* model (Eq. (8)) becomes mis-specified as one cannot divide both sides of Eq. (1) by population while keeping all other elements constant anymore. On the other hand, if the elasticity of P is estimated to be unity, then one can withdraw this latter factor (from Eq. (4)) by dividing both sides of the equation (Liddle, 2013a). One implication is that the dependent factor then fall into a population-weighted (per capita) metrics; thus collapsing the STIRPAT specification into a more standard *EKC* representation, as shown in Eq. (9). An answer was first brought by Liddle (2015) who collected data series for 26 OECD and 54 non-OECD countries with data spanning 1971–2011. Applying AMG and CCEMG estimators on a STIRPAT-type specification, findings showed that population elasticity does not significantly differ from unity (i.e., 0.97) for non-OECD economies; and created one of the first bridges between STIRPAT and *EKC/CEC/EEO* literatures. With a marginal elasticity of 0.57, the author failed to identify the same pattern for OECD countries, and shed light on the role of stage of development as potential confounder of the population-environmental impacts nexus.

5. State-of-the-art review

5.1. Survey of past empirical STIRPAT assessments: Multi- and single-country approaches

The STIRPAT literature combining data series with environmental impacts frameworks is rich and extensive. Over time, population and affluence elasticities have been recursively estimated within large panel assessments which often, encompass a wide range of structurally heterogeneous economies. Thus, case studies, selected data period, and econometric methodologies differ, whereas results remain conflicting. While all mentioned studies relied on the same theoretical architecture, the nature of the STIRPAT framework itself offer a wide flexibility to empirical modelers and econometricians (Lohwasser et al., 2020). For this reason, Vélez-Henao et al. (2019) underlined that estimates remain highly sensisitve to geographical locations differences and control variables choices. Also, it is worth noticing that the authors adopted contrasted approaches towards the treatment of the technology (T) variable, which cannot be neglected when analysing those divergences.

That is, despite those structural differences, a possibly-generalizable pattern emerges as a strand of this literature found larger population-related ecological elasticities with respect to per capita GDP (i.e., $EE_{IP} > EE_{IA}$). In general, multi-country investigations dominate the literature, whereas single-country analyses remain more incipient. This Section aims at separately presenting their main features. In Table 1, we first summarize the main information on multi-country STIRPAT panel assessments, and seek to offer an updated version of Liddle (2015). Followingly, Table 2 outlines single-country STIRPAT applications.

On the other hand, the recent years witnessed a gradual shift from multi-country panel assessments with estimates averaged at the sample level, to single-country analyses often based on the computation of series from multiple regions. Notwithstanding its relevance, this strand of the literature seems skewed towards China, which appears to be a dominant case studied. Another observation is that related econometric specifications present more frequent population-weighting (to regress per capita-transformed series on environmental change) while substituting P with an indicator for urbanization. Table 2 summarizes the main information on single-country STIRPAT investigations.

5.2. A note on STIRPAT studies' features and conclusions

While intense theoretical debates have shaped the STIRPAT literature over decades, associated syntheses and surveys failed to offer a common view.

In general, both population and income/affluence have been identified as significant drivers of carbon emissions by empirical studies employing a STIRPAT functional form. Nonetheless, population's elasticities seem exhibiting a larger impact than affluence in some cases (Cole and Neumayer, 2004; Liddle and Lung, 2010; Martínez-Zarzoso et al., 2007). On the one hand, Liddle (2013a) showed that much less consensus arises regarding the magnitude of population's elasticity since empirical conclusions seem to have diverged more than converged. It is clear that the STIRPAT literature has been refined through empirical engagements with various components of the model, and tested through extended econometric functional specifications. More flexible frameworks offered allowed a range of hypotheses testing using empirical data; including: (*i*) whether elasticities' magnitudes vary across development stages; (*ii*) whether population or affluence induce a larger marginal response from the environment; and (*iii*) whether population's elasticity is different from unity (i.e. whether population or impact/emissions grow at a faster pace than affluence). But on the other hand, Lohwasser et al. (2020) noticed that only a few studies, which generally focused on specific samples of countries or regional entities, underlined that per capita GDP elasticities exceed population ones. Given such a growing consensus regarding the sign and magnitudes exhibited by general ecological elasticities, the STIRPAT literature has progressively shifted from (re)-assessing the role of conventional factors (i.e., population and per capita GDP) to investigating parameters' sensitivities to the incorporation of additional control factors; with sometimes clear over-identification and noise variance bias. That is, since ample evidence that EE_{IP} is larger than EE_{IA}, many researchers suggested that environmental planners should instead orient their efforts towards internalizing the externalities induced by the first predictor (Lohwasser et al., 2020). For instance, Casey and Galor (2017) clearly highlighted that including demographic considerations within future policy planning is far from irrelevant. Indeed, population seems at the heart of global environmental change, but at the core of solutions: a 1% slower population growth rate is found to correlate with per capita income increases up to 7% with constant or marginally decreasing CO_2 levels. Finally, it is worth noticing that the STIRPAT literature conflicts with the findings from neighbouring research studies (i.e., which used indirect measures of CO_2 emissions such as energy demand; or examined the demand-side effect on environment-related metrics) conducted using non-STIRPAT frameworks. This is the case of Raftery et al. (2017) who elaborated a statistically-based probabilistic forecast of atmopheric polluting elements and suggested that demographic indicators fail to emerge as a substantially driving factors; which does not exclude these authors and others to suggest comprehensive consumption adjustements (Ahmed et al., 2019; Rauf et al., 2018; Shahbaz et al., 2017). Overall, some papers went one step further by offering nuanced discussions on A and P entities with refining potentials.

Regarding population (P), Jorgenson and Clark (2013) examined the varying global relationships among demographic sub-components and CO_2 releases. While population remains the widest contributing factor to carbon trends; the marginal elasticity of carbon emissions to population size turns much smaller in Africa. Hence, they showed that the magnitudes of the population coefficients are heterogeneous across regions, and may be confounded by initial stages of development. Also, York and Rosa (2012) developed a STIRPAT model to investigate the effects of population size on atmospheric pollutants, stratifying by total number of households and average household sizes. Thus, regressing the total number of households over atmospheric pollution gives a larger parameter when compared with average households' size because the emergence of an additional household is associated with a relatively high fixed capital, resource and environmental costs (transportation

Table 1
Summary of multi-country STIRPAT panel assessments.

Author(s)	Countries	Sample period	Method	Variables	$\widehat{\sigma}_{Aff.(A)}$	$\widehat{\sigma}_{Pop.(P)}$
Shi (2003)	88 countries	1975–1996	OLS	CO_2, YPC, POP, ISY, SSY	0.80	0.83(HI); 1.42(UMI); 1.97 (LMI); 1.58(LI)
Cole and Neumayer (2004)	86 countries	1975–1998	FDPR	CO_2, YPC, POP, AS, HS, E/Y, ISY, URB	0.89	0.98
Fan et al. (2006)	218 countries	1975–2000	OLS	CO_2, YPC, POP, AS, E/Y, URB	0.30(overall); 0.54(HI); 0.21(MI); 0.25(LMI); 0.33 (LI)	0.68(overall); 0.57(HI); 0.33(MI); 0.44(LMI); 0.26 (LI)
York (2007)	14 EU countries	1960–2000	OLS	CO_2, YPC, POP, SODPOP, URB	0.70	2.75
Martínez-Zarzoso et al. (2007)	23 EU countries	1975–1999	FDPR	CO_2, YPC, POP, E/Y, ISY	0.42(overall); 0.15(15 old EU countries); 0.34(8 new EU countries)	NA(overall); 0.71(15 old EU countries); 2.73(8 new EU countries)
York (2008)	15 former Soviet Republics	1992–2000	OLS	CO_2, YPC, POP, ADR, URB, SFDIY, MP	0.50	1.87
Liddle and Lung (2010)	17 OECD countries	1960–2005 (5-yr intervals)	OLS with 2WFE	CO_2, YPC, POP, AS, E/Y for I, SPECNF, URB	0.57	0.69
Jorgenson et al. (2010)	57 less developed countries	1990–2005	FDPR	CO_2, YPC, AS, ISY, URB, POPSLS	0.33	0.70
Jorgenson and Clark (2010)	22 developed and 64 less developed countries	1960–2005 (5-yr intervals)	FDPR	CO_2, POP, YPC, TSY, URB	0.65(overall); 0.95(OECD); 0.64(non-OECD)	1.43(overall); 1.65(OECD); 1.27(non-OECD)
Poumanyvong and Kaneko (2010)	33 high-, 43 middle- and 23 low-income countries	1975–2005	FDPR	CO_2, POP, YPC, E/Y, SSY, ISY, URB	1.08(overall); 0.83(HI); 0.99(MI); 2.50(LI)	1.12(overall); 1.12(HI); 1.23(MI); 1.75(LI)
Liddle (2011)	22 OECD countries	1960–2007	PCT, FMOLS	CO_2T, AS, REC, YPC	1.06	2.35
Martínez-Zarzoso and Maruotti (2011)	88 non-OECD countries	1975–2003	FDPR	CO_2, POP, E/Y, ISY, URB	0.72	NA
Menz and Welsch (2012)	26 OECD countries	1960–2005 (5-yr intervals)	FE	CO_2, YPC, POP, AS, BC, SECC, URB	0.36	0.78
Jorgenson and Clark (2012)	86 countries	1960–2005 (5-yr intervals)	PW-R with PCSE	CO_2, YPC, POP, URB, TSY	0.93	1.55
Zhu et al. (2012)	20 non-OECD countries	1992–2008	FE, SP	CO_2, YPC, EC, URB	1.12	0.79
Liddle (2013b)	23 high-, 25 middle- and 37 low-income countries	1971–2007	PCT, FMOLS	CO_2, URBD, YPC, POP, FE, FP, PTEC	1.00(overall); 0.44(HI); 1.00(MI); 0.97(LI)	1.05(overall); 1.05(overall; 0.68(HI); 0.86(MI); 0.70 (LI)
Knight et al. (2013)	29 OECD countries	1970–2007	FDPR	CO_2, POP, CEF, YPC, EMPS, AHW, EF, URB	0.59	2.25
Sadorsky (2014)	16 emerging countries	1971–2009	AMG	CO_2, YPC, POP, URB, E/Y	1.14(SR); 0.90(LR)	1.33(SR); 2.52(LR)
Liddle (2015)	26 OECD and 54 non-OECD countries	1971–2011	AMG, CCEMG	CO_2, YPC, POP, SPE, IEI	0.57(OECD); 0.97(non-OECD)	0.17(OECD); 0.20(non-OECD)
McGee et al. (2015)	173 countries	2001	OLS with FEs	CO_2, YPC, POP, URB, AS, SAL, TAL, DCN, DTC, SISA	0.60	0.609
Casey and Galor (2017)	156 countries	1950–2010	FE	CO_2, YPC, SWAC, URB, TSY	0.22	1.44
Ghazali and Ali (2019)	10 NICs	1991–2013	DCCE	CO_2, YPC, POP, LP, URB, TO, EINT, PCYRE	1.563	0.353
Koçak and Ulucak (2019)	19 OECD countries	2003–2015	P-OLS, FD-GMM, S-GMM	CO_2, YPC, URB, TSY, R&DwEA	0.105	0.525
Lohwasser et al. (2020)	84 countries	1980–2014	S-β-coefficients	EFP, YPC, URB, POP, SWAC	1.58	0.108
Gani (2021)	59 coal, 77 gas and 96 oil consuming countries	2002–2016	FE with PCSE	ENVQI, FC, YPC, SFTL, ASY, ISY, TSY, POPD, SE, GE, DIEC	0.0003(coal); 0.0002(gas); 0.0003(oil)	0.003(coal); −0.0009(gas); 0.002(oil)
Usman and Hammar (2021)	16 APEC	1990–2017	FGLS, AMG, CCEMG	EFP, TECI, FDI, YPC, RWEC, POP	0.5598	0.0017
Wu et al. (2021)	18 economies	2005	PDCT, 2WFE, GC-VECM	CO_2, POP, YPC, EC, RWEC, ISY, CO_2/FE	0.277	0.250
Boubellouta and Kusch-Brandt (2022)	30 EU countries	2008–2018	P-OLS, QRM	RECY, POP, YPC, Y/E, CRED, COLL	29.85	−0.02
Kazemzadeh et al. (2022)	19 emerging countries	2000–2016	KCT, OLS, PQR, MM-QR,	EFT, ECI, ICT, YPC, FEC, FR, TO, URB	0.67	−0.64(URB)

Source: Constructed by the authors.
Notes(a): CO_2: carbon dioxide emissions; CO_2T: carbon dioxide emissions from the transport sector; ADR: age dependency ratio; AHW: amount of hours worked; AS: age structure; ASY: share of agriculture in total GDP; BC: birth control; CEF: carbon energy efficiency; CRED: credit to private sector; COLL: amount of e-waste; DIEC: dummy coding for international environmental commitments; DCN: dummy variable for core nation; DTC: dummy variable for tropical countries; E/Y for I: energy-GDP ratio for the industrial sector; E/Y: energy-GDP ratio; EC: energy consumption; ECI: economic complexity; EF: energy efficiency; EFP: ecological footprint; EMPS: employment share; FE: fuel efficiency; ENVQI: environmental quality index; FP: fuel price; FR: fertility rate; IEI: industrial energy intensity; GE: government effectiveness; ICT: Information and Communication Technologies; ISY: industrial share over GDP; LP: labour productivity; MP: numnber of military personnel per 1000; POP: population size; PCYRE: Population carrying intensity of real economy; POPD: population density; POPSLS: share of population living in urban slums; PTEC: private transport energy consumption; REC: residential electricity consumption; RECY: e-waste recycling rate; R&DwEA: Research & Development with energy

applications; RWEC: renewable energy consumption; SAL: share of agricultural land over total land; SE: schooling enrollement; SECC: share of electricity consumption from coal; SISA: share of impervious surface area; SFDIY: share of Foreign Direct Investments in total GDP; SFTL: share of forests in total land; SODPOP: share of old dependent population; SPE: share of primary energy consumption from non-fossil fuels; SPECNF: share of primary energy consumption from non-fossil fuels; SSY: share of service in total GDP; SWAC: share of working age category; TAL: total agricultural land; TECI: technological innovations; TSY: trade share of GDP; URB: urbanization; URBD: urban density; Y/E: GDP-to-energy ratio; YPC: GDP per capita.
Notes(b): HI: high-income; MI: middle-income; LI: low-income; LMI: low- and middle-income; UMI: upper-middle income; 2WFE: two-way Fixed Effects (cross-section and time); AMG: Augmented Mean group estimator; CCEMG: Common Correlated Effects Mean Group estimator; DCCE: Dynamic Common Correlated Effects model; FD-GMM: First-Differenced-Generalized Method of Moments; FDPR: First-Difference Panel Regression; FE: Fixed Effects;FGLS: Fully-Generalized Least Squares; FMOLS: Fully Modified Ordinary Least Squares; GC-VECM: Granger Causality based on a Vector Error Correction Model; KCT: Kao Cointegration Test; MM-QR: Method of Moments Quantile Regression; PCSE: Panel-Corrected Standard Errors; PCT: Panel Cointegration test; PDCT: Pedroni Panel Cointegration test; P-OLS: Pooled Ordinary Least Squares; PW-R: Prais-Winsten; QRM: Quantile Regression Model; S-GMM: System-Generalized Method of Moments; SP: Semi-Parametric Model; S-β-coefficients: Standardized-β coefficients estimation. SR: short-run; LR: long-run; NA: non-assessed.

Table 2
Summary of single-country STIRPAT assessments.

Author(s)	Countries	Sample period	Method	Variables	$\widehat{\sigma}_{Aff.(A)}$	$\widehat{\sigma}_{Pop.(P)}$
Shahbaz et al. (2016)	Malaysia	1971Q1–2011Q4	BH-CC, ARDL, GC-VECM	CO_2, URB, YPC, KUPC, TO	1.08(LR); 0.0402(SR)	−12.39(URB-LR); −7.56(URB-SR)
S.-C. Xu et al. (2016)	29 Provinces in China	1995–2011	FGLS	CO_2, POP, Y/E, ISY	1.01	0.93
D. Liu and Xiao (2018)	30 Provinces in China	2000–2012	SDM	CO_2, POP, YPC, EINT, ES, ISY, TFI	1.156 (overall); 1.419 (Western); 0.917 (Central); 1.271 (Eastern)	1.069 (overall); 1.273 (Western); 0.768 (Central); 1.059 (Eastern)
Xiong et al. (2019)	Kazakhstan	1992–2004	RR	CO_2, POP, Y, SCC, TO, FDI, SIS, SSS, CO_2PUGDP	0.1079	2.0657
Zhang and Zhao (2019)	30 Provinces in China	1996–2005	S-GMM, FE-DK	CO_2, URB, YPC, E/Y, ISY, R%D	0.544	0.122(URB)
F. Xu et al. (2020)	China	2003–2015	PGC, FE	SO_2, IS(d), DG, IWW, URB, POP, YPC, ISY	0.001(for IWW specif.); 0.12(for SO_2 spef.); 0.174(for IS(d) specif.); 0.073 (for DG specif.)	0.454(for IWW specif.); 0.002(for SO_2 spef.); 0.986(for IS(d) specif.); 0.955(for DG specif.)
Huang et al. (2021)	China	FE-IV, GMM	CO_2, YPC, FEC, URB, FDI, EPAT, SSY	0.356	0.596(URB)	
Zheng et al. (2022)	30 Provinces in China	2001–2016	OLS, TRM	SO_2I, TPI, YPC, EINT, FDI, TEC, ISY, URB	−0.182	0.0853(URB)
C. Zhu et al. (2022)	China (construction sector)	2004–2017	RR	CO_2, BSY, EINT, TFP, LB, NVFACS	1.031 (unstandardized); 0.201 (standardized)	0.495 (unstandardized); 0.067 (standardized)

Source: Constructed by the authors.
Notes(a): CO_2: carbon dioxide emissions; SO_2I: carbon dioxide emissions intensity; CO_2PY: carbon dioxide emissions per unit of GDP; BSY: share of building sector in total GDP; DG: domestic garbage; EINT: energy intensity; ES: energy structure; EPAT: energy patents; FDI: foreign direct investments; FEC: fossil energy consumption; IS(d): industrial soot(dust); ISY: share of industrial sector in total GDP output; IWW: industrial wastewater; KUPC: capital use per capita; LB: labour; NVFACS: net value of fixed assets in the construction sector; POP: population size; SCC: share of coal consumption in total energy use; SO_2: sulfur dioxide emissions; SSY: share of services sector in total GDP output; TEC: total energy consumption; TFI: fixed-asset investment; TFP: total factor productivity (Obtained through a Cobb-Douglas production function); TO: trade openness; TPI: technological progress index; URB: urbanization; Y/E: GDP-to-energy ratio; YPC: GDP per capita.
Notes(b): ARDL: Autoregressive Distributed Lag Bound; BH-CC: Bayer-Hank Combined Cointegration; SDM: System Dynamics Model; FE: Fixed Effects model; FE-DK: Fixed Effects with Driscoll-Kraay standard errors; FE-IV: Fixed Effects with Instrumental Variables; FGLS: Fully Generalized Least Squares; GC-VECM: Granger Causality based on a Vector Error Correction Model; GMM: Generalized Method of Moments; OLS: Ordinary Least Squares; PGC: Panel Granger Causality test; RR: Ridge Regression model; S-GMM: System Generalized Method of Moments; TRM: Threshold Regression Model. SR: short-run; LR: long-run.

infrastructure and power and pipe facilities). Conversely, such inferences are less sensitive to a change in the number of individuals constituting the house. Those studies claim how nuanced population size effects can be, depending on the metrics employed. Another relevant element is that the spatial distribution of population is neither random, nor apolitical, and can thus not be neglected by future socio-environmental analyses (McGee et al., 2015).

Regarding the affluence (A) term, Jorgenson and Clark (2012) conducted a comparative study of the multiplicative impact of gross domestic product on CO_2 emissions; and assessed to which degrees nations' economic development help driving a"de-linking" effect from carbon emissions, or conversely, triggers this dynamic. They incorporated an interaction term between GDP per capita (logged) and each time point of the data (1960 being the reference t period) to isolate heterogeneities introduced by growth-specific time trend effects; but also accounted for the confounding effect of the historical stage of economic expansion on the GDP-CO_2 nexus by interacting per capita GDP with a binary dummy variable coding for the least developed countries of the sample. While atmospheric emissions appear to rise linearly over time with affluence (GDP), the strength of this interaction differs across countries. One observation is that advanced economies present a strong coupling dynamic between affluence and CO_2 levels until the end of 1970's, followed by a relative"de-linking" process through the early 1980s, to finally end with a relative steadiness from 2000's up to now. Conversely, emerging nations exhibit a strong and statistically time invariant coupling relationship, with much less differences in the sign and magnitude of correlation coefficients across periods.

6. Econometric challenges

This Section aims at providing an overview on econometric challenges that the STIRPAT literature attempted to address so far. Among them, one find the use of multicollinear vectors, the manifestation of reverse causality and simultaneity bias, the presence of a unit root in the data, cross-sectional dependence and spatial-correlation inferences among the series, heterogeneous slopes within the sample, non-linearities and asymmetries in the estimated responses, as well as non-standardized β-coefficients.

6.1. Multicollinear vectors, reverse causality and simultaneity bias

One striking observation from empirical STIRPAT applications is the bias of multicollinearity, which is often overlooked by studies. When uncontrolled, correlation inferences do increase the risk of misleading estimates because of the presence of one or several variables in the multivariate framework which can be linearly predicted from the others (Green, 1997). Moreover, additional identification concerns may arise EKC/CKC applications since slight measurement errors which, besides triggering endogeneity bias, may shift the"true" level of the estimated turning point or change the overall shape of the response function. As emphasized in Stern (2004), estimating the response of environmental change to transformed versions of a single variable (i.e., GDP, GDP^2, GDP^3) is likely to introduce correlation inferences, especially if no lagged variable is used. One old (but not oudated) illustrative example can be found in Grossman and Krueger (1991) who include the three year average of each (of the above) regressor (i.e., $\overline{GDP}_{it-}$, $\overline{GDP}^2_{it-}$, and $\overline{GDP}^3_{it-}$) within the same specification and capture the effect of past income levels in influencing current environmental change. One misleading issue is that most β_j ($j = 1 \ldots n$) are individually different from 0, but jointly highly significant. Thus, some of the conflicting results may not only come variation in the data, but also from the variety of uncontrolled multicollinearity scenarios and econometric permutations. Up to now, this methodological objection, which is as old as the empirical literature[6] is missed by most of the recent STRPAT papers while attempting to explain diverging estimates.

Reduced STIRPAT forms such as income driven inverted *U*-shaped representations (EKC/CKC) remain criticized for their theoretical caveats but also methodological and econometric robustness. As stated in Carson (2020)*"[…] assumption of a causal role of income growth and the inadequacy of reduced-form specifications that presume that a common income-related process […] adequately describes the generation of the pollutant of interest"* (p.5). Applying relatively less robust identification strategies when inferring causality on a public good like pollution is likely to generate bias. For instance, we do expect simultaneity (endogeneity) bias because income and pollution influence each other in a dynamic manner, as two-way mechanisms do operate (Lieb, 2003). On the one hand, it has been shown that exposure to atmospheric pollutants goes beyond general health consequences (work-day losses) (Van Ewijk and Van Wijnbergen, 1995), but also reduces the average cognitive performance, and thus productivity (Ebenstein et al., 2016; Roth, 2016). On the other hand, pollution adversely impacts forestry yields and harvest quantities, which in turn influence income trends. Naturally, a relevant question is whether the feedback effect of pollution on income prevail over the reciprocal causal link. On this point, Lieb (2003) surveyed the early literature and failed to identify evidence confirming the above-statement in previous macro-level studies.

Using a more micro-econometric lens, Ren and Tong (2008) offered an exhaustive discussion on the methodological challenges facing air pollution assessments. First, atmospheric particles differ by their characteristics and concentrations and are thus unlikely to be randomly assigned across places. Second, when both economic and environmental metrics evolve over time, often operate unobserved confounders. Third, wrangling localised pollution data is non-neutral in itself because the deployment and location of monitoring stations are neither random nor apolitical. Thus, uncontrolled spatial correlation, measurements errors, and misclassification bias are likely. For these reasons, Greenstone and Gayer (2009) suggested that the environmental pollution literature should shift towards the use of econometric settings which assume a zero-selection bias while controlling for endogeneity. One promising path is the utilization of quasi-experimental methods (i.e., Instrumental Variables (IVs), Difference-in-Differences (DiD) and Regression Discontinuity Design (RDD)), which exploit natural reforms to identify and isolate exogenous changes in the variable of interest to be estimated, under the condition that strict assumptions are fulfilled.[7] Nonetheless, because quasi-experiments postulate unit-level treatment assignment and no spillover effects infering across them, Deschenes and Meng (2018) pushed the frontier of the literature forward and proposed to model distinct unit and group level exposures to pollutants and compare the mean outputs across units being the source, and those being not the source of atmospheric pollution.

6.2. Non-stationary data exhibiting unit root processes

First, Liddle (2015) showed that variables employed in STIRPAT assessments correspond to stock (i.e., population or physical capital), or stock-based indicators (i.e., GDP, emissions, and energy use themselves driven by stocks). That said, the graphic representation of those series over time embed significant trends and cycles, which suggests the possible presence of a unit root, and thus non-stationary series. This statement is supported by a strand of the energy-GDP-CO_2 nexus literature which frequently pointed out that GDP per capita, energy use and carbon emissions time-series fail to reject the $H0$ hypothesis of non-stationarity in levels (Magazzino and Schneider, 2020; Shahbaz et al., 2013; Simionescu and Schneider, 2022). However, when used in first-difference levels, those series present significant integration properties (of order 1 - $I(1)$[8]).

When performed on non-stationary data, time-series regression models (i.e., OLS, FE, and RE) may generate spurious estimates, with unreliable general statistics (R-squared, t-statistics) (Beck, 2008; Kao, 1999). Nonetheless, it is striking to observe the use of non-treated annual data series is a range of first-generation STIRPAT analyses.[9] For instance, Cole and Neumayer (2004) stressed that slightly larger population elasticities reported by Shi (2003) could be due to the use of data series containing a unit root. On the other hand, it is true that the use of first-difference-transformed series turned to be a frequent tool through which empirical studies would control for non-stationarity. While it is clear that first-differencing offers relevant statistical advantages (i.e., it is often sufficient to remove a potential unit root), it does also modifies the general structure of the model, from long- to short-run implications (Liddle, 2015). Assuming a fully-first-difference-

[6] See Harbaugh et al. (2000) who were among the first to invoke the multicollinearity argument to explain differences in estimated turning points of Grossman and Krueger (1991) (6151 PPP$) and Barrett and Graddy (2000) (7286 PPP$).

[7] In RDD settings, individuals do not control the assignment variable τ (Lee and Lemieux, 2010). Regarding DiD, the absence of any treatment should enable parallel trends among groups (Abadie, 2005). Concerning IVs, the instrument must be correlated with the treatment but orthogonal to the unobserved determinants of the outcome (Angrist and Imbens, 1995).

[8] A variable can be said to be integrated of order d (denoted $I(d)$), if and only if it must be differenced d times to become stationary. Accordingly, a stationary variable is said to be integrated of order zero (i.e., $I(0)$), whereas a variable that must be differenced only once to reject the null hypothesis of non-stationarity is considered as integrated of order one or $I(1)$ (Bradford et al., 2005)

[9] See Shi (2003); Fan et al. (2006); York (2007); and York (2008) - those studies are also relevantly summarized in Liddle (2015).

transformed model, one reason is that estimated parameters capture the extent to which a unity % change in the growth rate of an independant explanatory variable induces a percentage marginal change in the rate of increase of the dependent element, instead of strict percentage change in the absolute value of each series.

Similarly, Stern (2010) stressed that such statistical concern applies to EKC assessments which relied on non-linear transformations of non-stationary variables to estimate the potential turning point $-\beta_1/2\beta_2$. When the presence of a unit root is effectively controlled, Wagner (2008) underlined that the GDP-CO_2 emissions (per capita) response function may exhibit a positively monotonic shape rather than an inverted *U* one. Overall, it is worth noticing that both recent CKC and energy-GDP nexus literatures estimated long-run elasticites through stepwise causality testing frameworks accounting for non-stationary series (Iwata et al., 2012).

6.3. Cross-sectional dependence and spatial correlation among countries' series

Over the recent decade, the Time-Series Cross-Section (TSCS) econometric literature showed a keen interest for cross-sectional dependence-related bias. From an empirical standpoint, it is clear that inter-dependencies driven by the economic and financial integration of markets, and taking the form of spatial correlations across countries cannot longer be neglected by estimation models (De Hoyos and Sarafidis, 2006). In a globalized context (trade and financial), regional and national linkages operate through common macroeconomic shocks affecting multiple entities; common international policy directions pushed by inter-state organizations (i.e., the Bank for International Settlements (BIS), the Kyoto Protocol and the Paris Agreement, International Labour Organization (ILO)); spillover effects and technology transfers across sectors belonging to different industries and countries (Liddle, 2015).

From a statistical point of view, a cross-sectionally dependent data structure can emerge in the presence of common shocks, spatial dependence and unobserved components which are captured by the error term (De Hoyos and Sarafidis, 2006). However, neither Random and Fixed Effects model (RE and FE), nor the Mean Group (MG) estimator from Pesaran and Smith (1995) are longer robust when cross-sectional dependence arises between panel members (Eberhardt, 2012; Kapetanios et al., 2011). When left unturned, those stones introduced inconsistent estimates and spurious inferences (Andrews, 2005; Eberhardt and Bond, 2009; Weinhold et al., 1999). Here, we stress that appropriate estimation models identifying and accounting for those inferences should be employed.[10] Besides addressing this issue, the Pesaran (2006) Common Correlated Effects Mean Group (CCEMG) estimator accounts for time-variant unobservable with heterogeneous effects across panel members and thus solves mis-identification problems (Eberhardt, 2012). Compared to the Mean Group estimator from Pesaran and Smith (1995), the CCEMG model controls for unobservables and potential endogeneity derived from omitted variable bias. This estimator is developed by extending a conventional Pooled Fixed Effects (PFE) regression function using the cross-sectional arithmetic averages of the dependent and *i* independent factors. Thus, these cross-section averaged regressors are included as additional covariate in each of the *N* panel specification to capture cross-section dependence dynamics, before calculating the averages of individual-level estimates. As pointed out in Liddle (2015), except in Jorgenson and Clark (2012), the early literature failed to estimate long-run elasticities using methods robust to cross-sectional correlation (Cole and Neumayer, 2004; Martínez-Zarzoso et al., 2007; Shi, 2003; York, 2007, 2008). Even in more recent EKC/CKC studies, controlling for this bias remains unsystematic, whereas even the most recent energy-GDP nexus analyses exhibits the same features (J.-L. Liu et al., 2020; Pata and Aydin, 2020; Radmehr et al., 2021; Saidi and Omri, 2020). Nevertheless, some relevant exceptions remain (Liddle, 2015; Usman and Hammar, 2021), but are insufficient to erode our general rule of observations.

6.4. Heterogeneous slope coefficients, non-linearities and asymmetries

In panel assessments, it is not unlikely to face issues linked to country-specific heterogeneities. While we do have evidence that considerable cross-country structural differences do occur in terms of income, industrial organization, technological integration, resource abundance and energy policies, a range of previous STIRPAT assessments estimated a unique vector of parameters averaged at the sample level using pooled regressors. Other STIRPAT studies splitted the panel according to a single criterion (i.e., often along income lines with arbitrary threshold levels) before running group-level regressions (Poumanyvong and Kaneko, 2010). But in general, we stress that omitting to control for these inferences is also likely to bias the estimated responsiveness of CO_2 emissions to endogenous changes in a set of anthropogenic factors, because it is highly unlikely that population- and affluence-environment response functions have homogeneous shapes across countries. Thus, compiling panel of countries displaying heterogeneous stages of development generate estimates that are hardly generalizable, with very few policy applications and implications for single study cases. Said in other words, substantial differences in individual STIRPAT elasticities do exist and cannot longer be overlooked. For instance, Deacon and Norman (2006) employed non-parametric methods and suggested that country-level estimated parameters do not align with panel patterns. This is the case of RE and FE models. Besides failing to detect heterogeneous slopes,[11] they estimate parameters by pooling the country-specific series all-together. However, under heterogeneous slopes across panel members, those conventional approaches generate inconsistent standard errors (Blackburne III and Frank, 2007). To deal with this bias, recent STIRPAT studies progressively shifted towards long-run estimation methods allowing for high-degrees of within-panel heterogeneities (Liddle, 2011, 2013a). Similarly, the use of long-run panel estimators accounting for heterogeneous slope coefficients across countries in the CKC and EEO literatures does exist (Apergis and Payne, 2009, 2010b; Lean and Smyth, 2010) but is not systematic (Iwata et al., 2011; Iwata et al., 2012). For instance, the two-steps Pesaran and Smith (1995) Mean Group (MG) model incorporates $N-1$ year dummies to capture common dynamic effects and construct a pooled regression specification, before performing the estimation on a first-difference OLS framework (*i*). Then, the coefficients of the year dummies $\widehat{\gamma}_t$ capturing common dynamic processes, are incorporated within the group-specific regression equation with an intercept to capture time-invariant fixed effects (*ii*). Besides controlling for unobservables, this approach has the advantage to estimate a different vector of parameters (i.e., intercept, slope coefficient and error variance) for each panel member with efficient standard normal errors.

EKC/CKC studies postulated that the shape of the income-emissions

[10] The literature commonly uses the Frees' test of cross sectional independence (Frees, 1995); the Friedman's test of cross-sectional independence (Friedman, 1937); the Breusch Pagan LM test of independence (Breusch and Pagan, 1980); or the more recently developed Pesaran (2004) Cross-section Dependence (CD) test. For a complete discussion on the respective properties of cross-sectional dependence tests, see De Hoyos and Sarafidis (2006).

[11] While starting from the conventional baseline of the Swamy's test (Swamy, 1970) based on small $N < T$, Pesaran and Yamagata (2008) proposed a slope homogeneity testing method for panel data displaying larger *N* and *T* properties. The null of homogeneous slope coefficients, indicating that all slope coefficients do not differ across cross-sectional units, against the alternative of heterogeneity. One competitive feature of this procedure is that it automatically assumes a vector of heterogeneous constants. See Breitung and Salish (2021) for an up-to-date discussion on this topic.

response function may change *within* and *across* income levels. Conventionally, this has been addressed by including a squared version of the income factor within the specification and test whether the estimated parameter associated with this variable is statistically significant and negative. In this case, this indicates evidence of that an inverted *U* shaped curve among explanatory and dependent variables holds. But once again, relying on non-linear transformations of per capita GDP variables that are not integrated of order 0 ot 1 (i.e., $I(0)$ or $I(1)$) may generate misleading turning point estimates of $-\beta_1/2\beta_2$ (Stern, 2010). Furthermore, Liddle (2015) argued that polynomial equations do not allow for positive elasticities differing across development levels. To overcome this issue, Liddle (2013a) motivated the use of panels stratified by income groups coupled with models accounting for heterogeneous slopes to avoid such spurious nonlinear transformations and control for heterogeneities, both *across* and *within* the income groups of economies. While further examined below, we recommend exploiting models enabling heterogeneous slope coefficients to identify a potential non-linear pattern in GDper capita's or population's impacts. Finally, Itkonen (2012) underlined that the non-linear structure of the CKC baseline (regardless the nature of the stationary properties of the series) is not compatible with Vector Auto-Regressive (VAR) estimators, although they are widely used in the emissions-energy-output (EEO) literature. Thus, this highlights how VAR-based approaches combined with transformed series may still produce misleading outputs.

This is in line with the main conclusions of energy-GDP nexus studies when revealing that income elasticities to energy use differ with development levels. Thus, countries with comparable structural features should adopt similar energy policies (Dagher and Yacoubian, 2012). Some researchers went one step further by conducting a large panel investigation over various income groups, but limited the scope of their analysis to electricity use. For instance, Apergis and Payne (2011) categorized 88 economies into four different samples following the World Bank income group classification. The authors found that estimates for low-income economies validate the *"growth hypothesis"*, whereas as income per capita increases, inter-dependencies among power use and industrial development may shift towards a feedback pattern, so that any fluctuation/shock in GDP could be directly transmitted back to power consumption levels, and vice versa. Such observation corroborates Bildirici and Kayıkçı (2012), who stressed that the electricity-GDP hypotheses do not differ across income groups belonging to Commonwealth of Independent States (CIS) countries. Nonetheless, this theory remains eroded by Akinlo (2008) and Acaravci and Ozturk (2010) who underlined that energy-GDP nexus findings cannot be generalized to development groups in 11 sub-Sahara African economies and 19 developed European countries, respectively. Recently, by estimating optimal thresholds of energy intensities, above which energy use non-negligibly jeopardizes income, Aydin and Esen (2018) shed light on another potential confounder of the energy-growth relationship. All in all, knowing that the human-environment relationship can be non-linear and asymmetrically respond to negative or positive shocks, researchers face the challenge that GDP and energy consumption levels as well as population size, may simultaneously play out when estimating environmental change.

6.5. Parameters' juxtaposition bias: Potential for univariate and partial β-standardizations

In their non-standardized form, ecological elasticities capture the shape of the relation between a predictor variable, and its associated environmental response. Hence, those are said *predictive* as they estimate an approximate arithmetically averaged response (i.e., percentage variation) linked with a unity % change in the predictor factor (Lohwasser et al., 2020). When samples are large enough and data series span a relatively long time period, those coefficients are likely to stay stable across case studies, sub-samples, and regions, thus ensuring comparisons with minimized bias. In addition, the conventional logarithmic transformation of the STIRPAT equation (see Eq. (5)) lowers the skewness of the series by shifting outliers adjacent to the dimension of the panel, while normalizing kurtotic distributions of continuous variables. Also, it partially allows heterogeneous variances and raw units in the predictor variables, and finally ensures percentage(%)-based predictions (Lohwasser et al., 2020).

However, Lohwasser et al. (2020) demonstrated that an important econometric challenge emerges when predictor variables follow different trends and dynamics, whereas the main interest is to estimate and interpret their relative individual role with respect to the variable of environmental change. In this context, a bias may arise due to the fact that units and variance may significantly differ. Hence, calculating ecological elasticities should be complemented with a β-standardization of regression coefficients following the methodologies proposed in Bring (1994), Schielzeth (2010), and Germán-Soto and Gutiérrez Flores (2015). In the general econometric literature, standardization aims at weighting the past variation of variables using their respective standard deviations such that: $\beta_x^S = \beta_x \cdot \frac{\sigma_x}{\sigma_z}$; where β_x^S refers to the standardized beta-coefficient of the independent variable x; β_x corresponds to the estimated parameter of any explanatory independent x; σ_x is the standard deviation of x whereas σ_z indicates the standard deviation of the dependent variable z (Wooldridge, 2015). Once standardized, estimated coefficients capture the variation of the response variable expressed in standard deviations metrics following a unity standard deviation shift in a given predictor factor, holding other independent explanatory drivers constant (ceteris paribus) (Lohwasser et al., 2020). In the same way, the squared-transformation of the standardized coefficient associated with a given independent variable captures its relative input to the the response factor's general variance, and thus gives information on the relative influence of each individual driver[12] (Lohwasser et al., 2020). A relevant non-STIRPAT-related application of such standardization method can be found in Germán-Soto and Gutiérrez Flores (2015), who analysed the determinants of regional patents activity in a panel of Mexican states.

Nonetheless, Lohwasser et al. (2020) relevantly stressed how standardization techniques remain unknown or incipient to the STIRPAT tradition. While extensively used in multiple regression models, it is striking to observe how weighting the past variation of identified anthropogenic drivers using their respective standard deviations is far from systematic in the STIRPAT literature, and in most cases, absent. This indicates evidence of a critical lack since standardization may sharply improve comparison and juxtaposition potentials of each predictors' impact on environmental change. Another reason is that a unit unit change in a given variable (i.e., per capita GDP) may not be equivalent to a unit change in another driver (i.e., population) because of standard deviation differences. Converting the interpretation mechanisms from percentage change to standard deviation units accounts for the scale of changeability of each variable. Compared to standard ecological elasticities, this is thought to ensure a better juxtaposition of parameters with more valuable additional information for policy purpose. However, Lohwasser et al. (2020) mentioned two remaining limits to comparability:

- β-coefficients remain sample- and time-period-specific. Thus, study-related heterogeneities make cross-results comparison challenging (King, 1986) (*i*).
- Interaction dynamics among predictors may modify the true effect of one individual variable due to combined statistical inferences. Hence, Schielzeth (2010) showed that the larger is the correlation among drivers, the higher is the joint contribution and the lower is a variable's respective impact. This is particularly relevant for

[12] Notice that in most cases, summing squared-transformed standardized parameters is lower then the total variance of the response term due to a non-negligible effect of $var[\varepsilon]$ (the error term) on $var[z]$.

multivariate STIRPAT frameworks involving three or more predictors. Conversely, β-coefficients derived from standard trivariate IRPAT models (with two individual predictors) are less subject to this problem, although both prediction factors may hide some remaining confounding inferences (Thayer, 1991) (*ii*).

While the first limit is a recursive problem for a literature in search of general principles, the second issue can be addressed using partial standard deviations (instead of ordinary standard deviations). This method can be traced back to Bring (1994) who first demonstrated that standardized coefficients' calculation could be extended with a Variance Inflation Factor (VIF) to account for potential multicollinearities among predictor variables; such that: $VIF_x = \frac{1}{1-R^2_{k-1}}$; where k refers to the number of covariates; R^2_{k-1} indicates the determination coefficient for a prediction factor x that is regressed on $k-1$ terms. Thus, Schielzeth (2010) re-stated that partial standard deviation $\sigma_x^{partial}$ can be calculated as: $\sigma_x^{partial} = \frac{\sigma_x}{\sqrt{VIF_x}}.\sqrt{\frac{n-1}{n-k}}$; where n encompasses all consecutive observations; which allows the calculation of $\sigma_x^{partial}$-based standardized coefficients $\beta_x^S = \beta_x.\frac{\sigma_x^{partial}}{\sigma_z}$. Compared to ordinary β-coefficients, partially standardized parameters presents the advantageous feature of directly capturing the individual contribution of a predictor to the change recorded in the response environmental variable (Bring, 1994). There is no doubt that, in the STIRPAT context where equation specifications exhibit three- or more individual predictors (A, P, and a combination of additional controls contained in T), β-coefficients based on partial standard deviations present a high potential for shifting the literature from ecological elasticities' interpretation which exclude the scale of changeability of each variable to estimating means of standard deviations of log-transformed data with more reliable parameters' juxtaposition. As emphasized in Sub-Section 6.1, most STIRPAT analyses relied on data series integrated of order 1 ($I(1)$), which implies interpreting coefficients based on first-differenced-transformed regressors free of unit root. While there seems to be a trade-off between relying on original but less comparable ecological elasticities on the one hand, and more reliable but statistically transformed coefficients on the other hand, we concur with Lohwasser et al. (2020) when stating that standardization processes should complement non-standardized EE, especially if the explicit aim of the study stands in understanding the analogous weight of individual explanatory factors in predicting the response variable. This thus become more systematic in empirical STIRPAT analyses.

7. Prospect for future research: Directions for theory and empirics

Notwithstanding the above econometric challenges, STIRPAT models keep leading a promising and fruitful research direction. This Section aims at highlighting potential prospects for future studies, with both theoretical and empirical takeaways.

From a theoretical standpoint, York et al. (2003) largely demonstrated the analytical power of STIRPAT and ecological elasticites tools. They were the first to shed light on the paradoxical phenomenon, so far missed by studies employing other methods (including IPAT and ImPACT models), that affluence displays a marginal decreasing CO_2 emissions effects whereas energy footprints rises more than proportionally with per capita GDP. They indicated that previous *decarbonization* interpretations were misleading, because this"progress" seems coming at the expense of other environment-related metrics. Besides the destruction of biodiversity and ecosystems through hydroelectric dam and nuclear power plants development, low-carbon energy deployment is changing the nature of the demand for resource but fails to disconnect from global material use and solid waste generation trends (Magazzino et al., 2020). However, this reverse side of the coin remains overlooked in may recent studies, thus highlighting critical caveats in the literature. Future study should incorporate those environmental externalities that are spatially and timely correlated with a decline in the carbon intensity of per capita GDP. Undoubtedly, this research direction cannot go without comprehensive progress in locating, identifying, and quantifying a larger set of environmental damages, including habitat losses, chemicals concentration in soils and water, as well as species migrations induced by waste landfills. In this regard, knowledge derived from the urban ecology literature may help reconcile theoretical STIRPAT frameworks with empirical analyses together.

Empirically, it is worth noticing that typical STIRPAT panel-based approaches employ data spanning from 30 to 50 consecutive years, with sometimes missing values for recent series, always absent information before 1960, and estimate elasticities averaged at the sample level (all countries being gathered together). Except when the Pesaran (2006) CCEMG model is performed, countries' series are individually collected, but gathered together. However, Liddle (2013a) highlighted that those studies relied on the constraining assumption that current high-income economies, when located at some incipient stages of their industrial process, behaved similarly to low-income countries today. Yet, we do already know that macroeconomic behaviours and trends are unlikely to survive the diversity of historical trajectories, and the narrative of time. On the one hand, looking at the World Economy Historical Statistics (WEHS) from Maddison et al. (1995) shows that low-income countries like Burkina Faso, Honduras, and Pakistan display today's per capita GDP comparable to Western early industrialized countries in 1820. On the other hand, it appears limited to support that those economies present identical industrial strategies, or that they share similar prospects for their future developments. Through technological consumption patterns diffusion as well as free-trade agreements and transport development, *"latecomers"*, as designated in Gerschenkron (2015) took a few decades to achieve major development steps which required a century to current industrialized countries. If data availability allows that, future empirical STIRPAT investigations should rely on historical series encompassing the largest development trajectory. This would not only bring more nuanced ecological elasticities, but also stress how the sign and magnitude of those parameters may vary with identified stages of global development (such as those defined in Rostow (1959) (i.e., traditional society (*i*), preconditions to take-off (*ii*), take-off (*iii*), drive to maturity (*iv*) and age of high mass consumption (*v*))).

While policy implications remain based on context- and region-specific features, some helpful tools are available for a STIRPAT literature in search of general principles. Nowhere on the planet, environmental changes induced by population and per capita GDP trajectories can be disconnected from profit accumulation systems based on resource extraction and transformation on the supply side, and growing energy-intensive needs for consumption and combustion purposes on the demand side. As stated in Sub-Section 6.4, better identifying the relative importance of each variable in forecasting the environmental response variable through univariate and partial β-standardization methods could offer a better comparability potential to the results. However, there is a need to investigate the statistical limits of standardization processes (notably in terms of interpretation of the parameters which are no more general elasticities), and how do those apply to STIRPAT-derived estimated coefficients (Lohwasser et al., 2020). A first stone has been laid in Schielzeth (2010) when critically discussing the use of simple arithmetic average to improve the interpretation potential of estimated parameters but much more should be done.

So far, the stochastic validity of this framework has been extensively tested using multi-country panel approaches, and sometimes at the national scale with provincial information. Nonetheless, much less attention has been drawn to local level-investigations. Said in other words, there is a need to further understand whether STIRPAT models successfully capture and describe the human-economy-environment relationship at *all* levels, including county and city scales. A seminal stone has been laid in Roberts (2011) who attempted to combine the first STIRPAT framework with a post-Fordist landscape. While estimates

derived from a traditional econometric framework turned inconclusive, the author raised the hypothesis that there may exist a scale-study limit (agent or household level) after which no relevant information can be extracted from data series. Although individuals do present the ability to implement behavioral changes that can reduce some types of local environmental damage, there are many other environmental externalities on which micro-scaled decision-making processes turns negligible. One reason may come from the perpetual reallocation of resources extraction, producing plants and consumer locations (i.e., shifts in the economic landscape) induced by sectoral and sociological structural changes, trade and financial integrations and international competition. But in general, local-based evidence are lacking and restrain our understanding of the complex human-environment relationship to global scaled estimates and policies, thus avoiding the range of potential solutions which could arise from municipal commitments.

8. Concluding remarks

While physical and natural scientists have long developed sophisticated tools to understand biogeochemical processes, questions remained on how to identify, isolate and deliver unbiased measures which accurately shape the anthropogenic-global environmental change response function. IPAT, ImPACT, and STIRPAT baselines arose as relevant candidates as they successfully combined theoretical extensions (e.g., ecological elasticities, technology (*T*) term decomposition, Carbon Kuznets Curve functional form) with extensive empirical assessments using real data series. But the STIRPAT model clearly moved our approach towards the driving factors of environmental impact forward by meeting simultaneously the three criteria defined early in York et al. (2003): STIRPAT allows efficiency or intensity assessments (*i*); STIRPAT provides a framework to infer causality (*ii*); STIRPAT enables forecasts of future effects (*iii*). Stratified in two strands (multi- and single-country approaches), this literature nonetheless failed to generate a strict consensus with respect to the magnitude of population and affluence elasticities, although some general principles seem to have emerged.

On the one hand, those divergences can be linked to the variety of models' specifications and control variables' choices reported, as well as the use of different proxies for the *T* term. On the other hand, past STIRPAT analyses attempted to address econometric challenges in diverse manners. Upon them, identification strategies which exclude multicollinear vectors, reverse causality and simultaneity bias, the use of non-stationary data exhibiting a unit root; uncontrolled cross-sectional dependence and spatial correlation among the series; heterogeneous slopes, non-linearities, asymmetries inferences; as well as a misleading understanding of the relative weight of individual explanatory variables in predicting the response variable due to unstandardized coefficients; contributed to divide the STIRPAT literature. To reconcile empirical findings, and unlock the policy potential of STIRPAT frameworks in unveiling competitive human-environment dynamics, key research directions emerge for both theory and empirics. Upon them, we recommend future studies to incorporate environmental externalities that are spatially and timely correlated with a decline in the carbon intensity of per capita GDP; rely on historical series encompassing the largest development trajectory (this would not only bring more nuanced ecological elasticities, but also stress how the sign and magnitude of those parameters may vary with identified stages of global development); assess the statistical limits of standardization processes (notably in terms of interpretation of the parameters which can no more be considered as general elasticities), and how do those apply to STIRPAT-derived estimated coefficients; examine the potential of local-based estimates to enrich our understanding of the complex human-environment relationship with non-global scaled evidence. Undoubtedly, there is left for promising future research.

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Nicolas Schneider: Writing – original draft, Investigation, Resources, Methodology, Formal analysis, Validation, Conceptualization, Writing – review & editing.

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